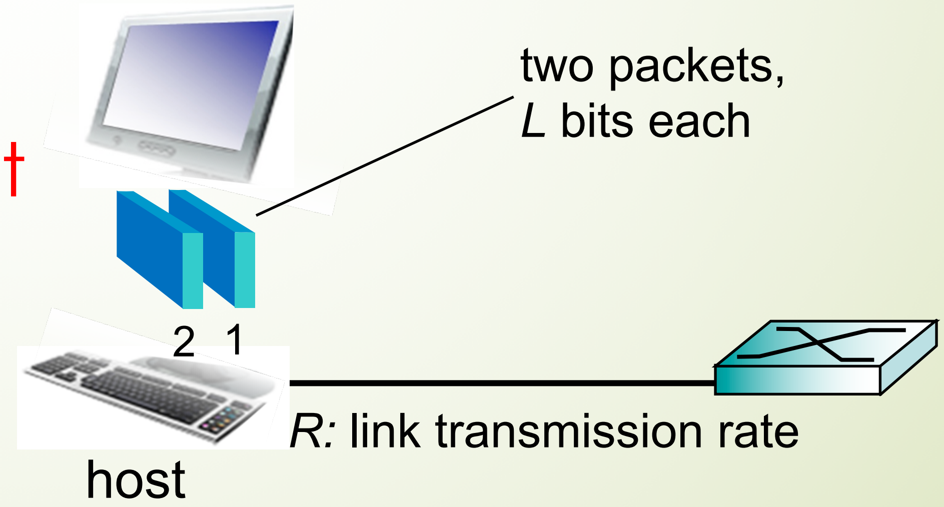
Host's Sending Function

- takes application message
- breaks into smaller chunks, known as packets, of length L bits
- transmits packet into access network at transmission rate R
  - link transmission rate = link capacity = link bandwidth

Q: What is the packet transmission delay?

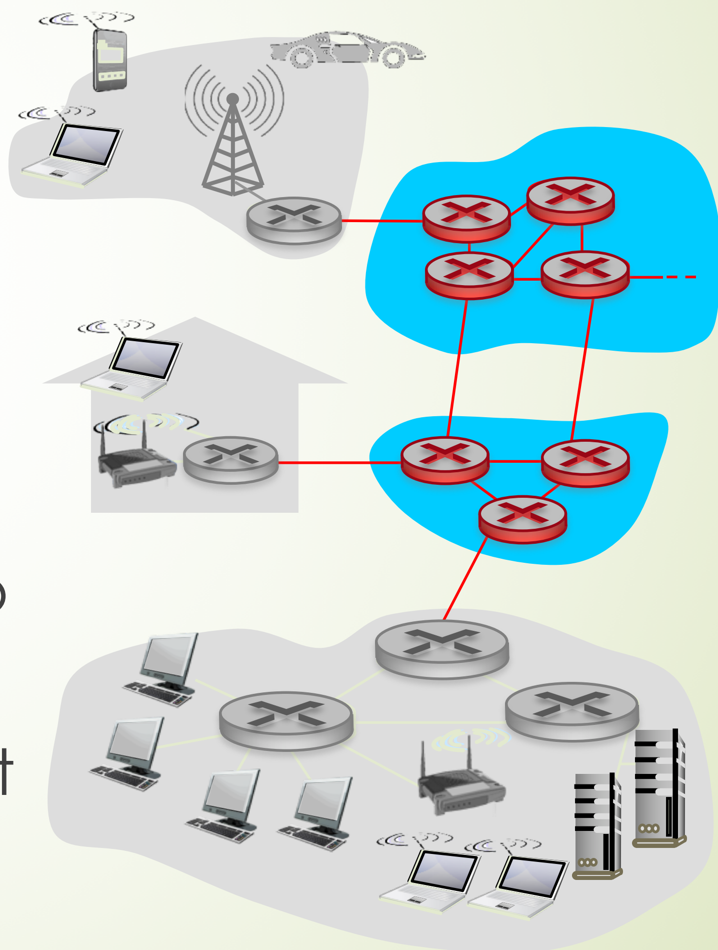


Agenda

- What is the Internet
- Network Edge
- Network Core: Circuit Switched vs. Packet Switched
- Protocol Layers and UNIX Sockets
- History and Internet design philosophy

Network Core and Packet Switching

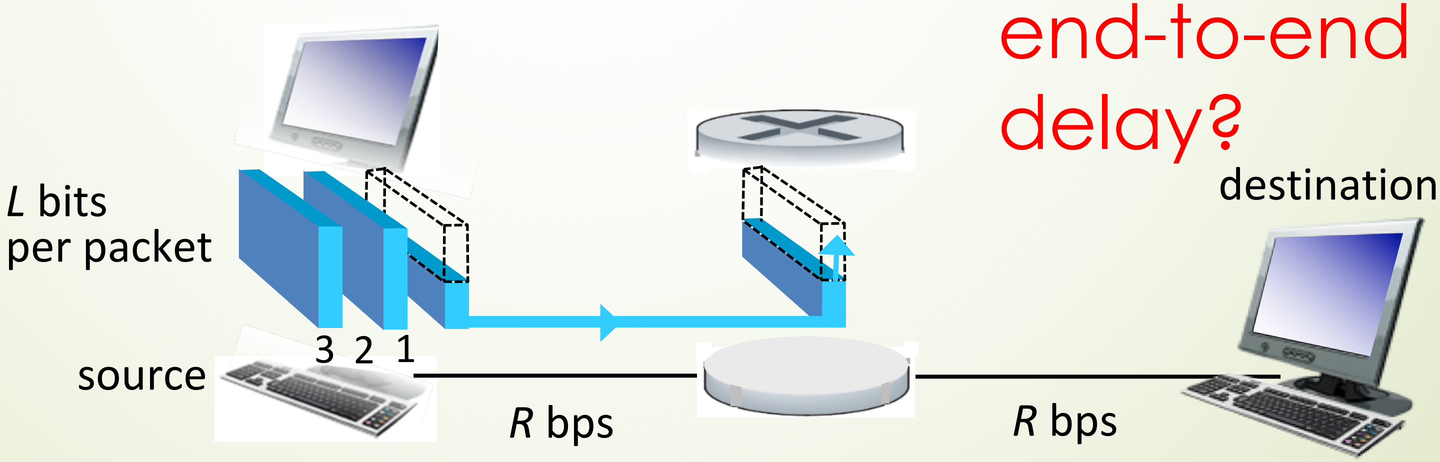
- interconnected routers
- packet-switching: hosts break application-layer messages into packets
  - forward packets from one router to the next, across links on path from source to destination
  - each packet transmitted at full link capacity



Store-and-Forwarding Principle

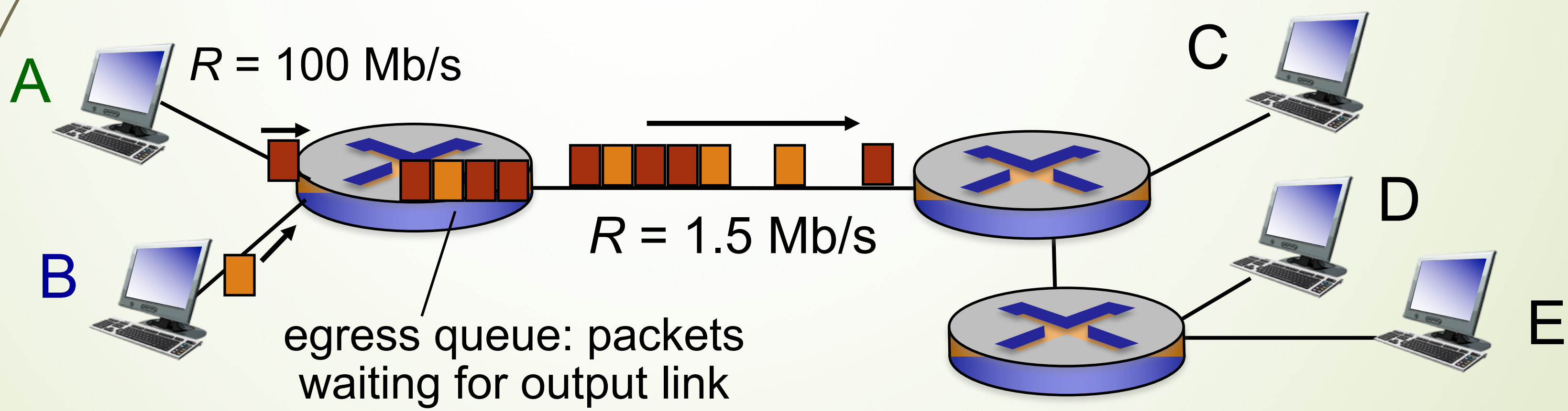
- takes *L/R* seconds to transmit (push out) *L*-bit packet into link at *R* bps
- *store and forward:* entire packet must arrive at router before it can be transmitted on next link

Q: What is the end-to-end delay?



Queueing Delay and Packet Loss

- **Q: Why does packet switching suffer from queueing delay and packet loss?**

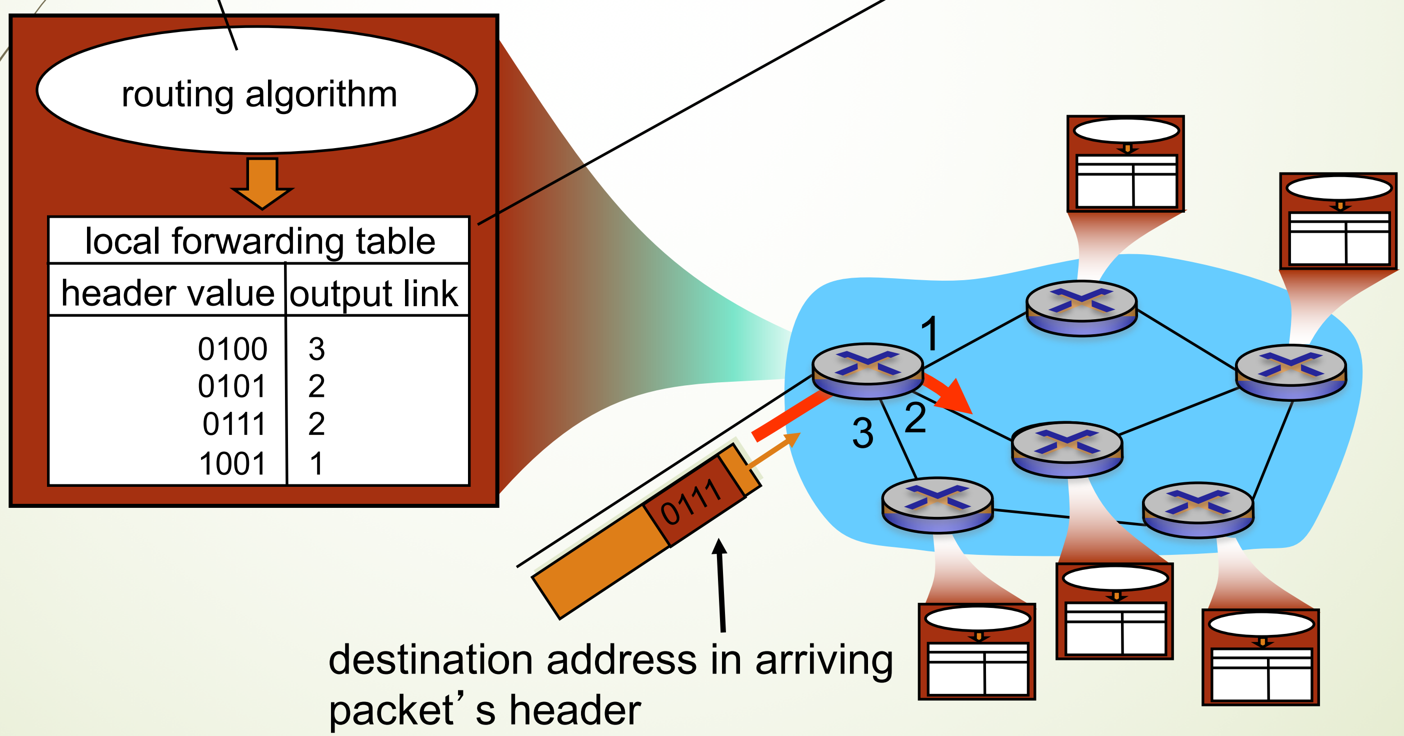


Two Main Functions of Routers

routing: **determines source-destination route taken by packets**

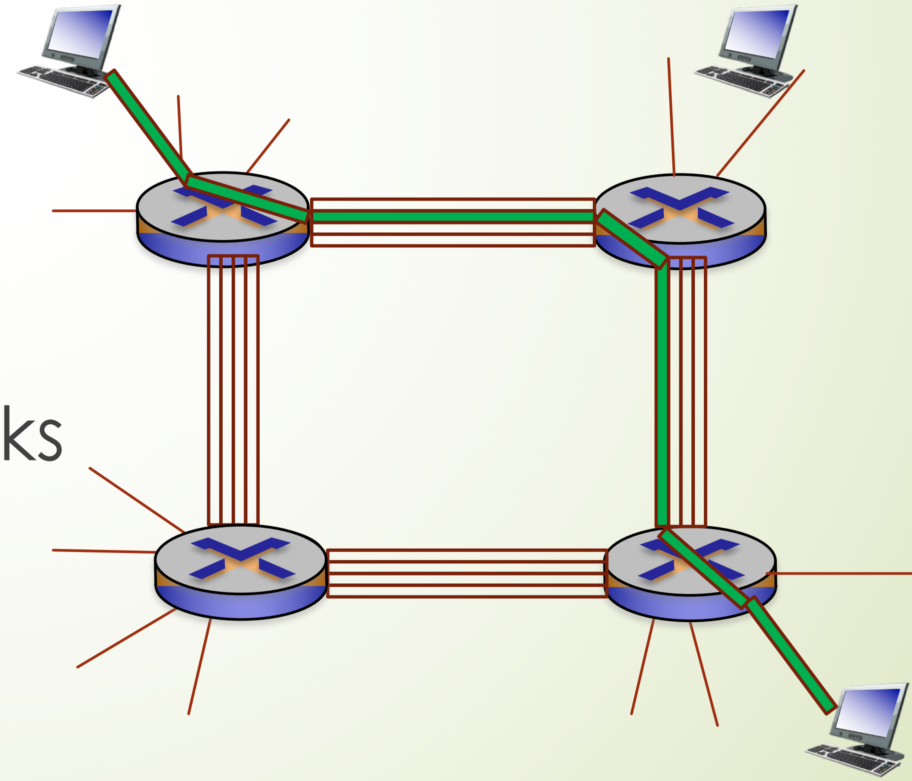
- ***routing algorithms***

forwarding: move packets from router's input to appropriate router output



Circuit Switching

- end-end resources allocated to, reserved for "call" between source & dest:
- dedicated (guaranteed) resources: no sharing
- circuit segment idle if not used by call: no sharing
- telephone networks

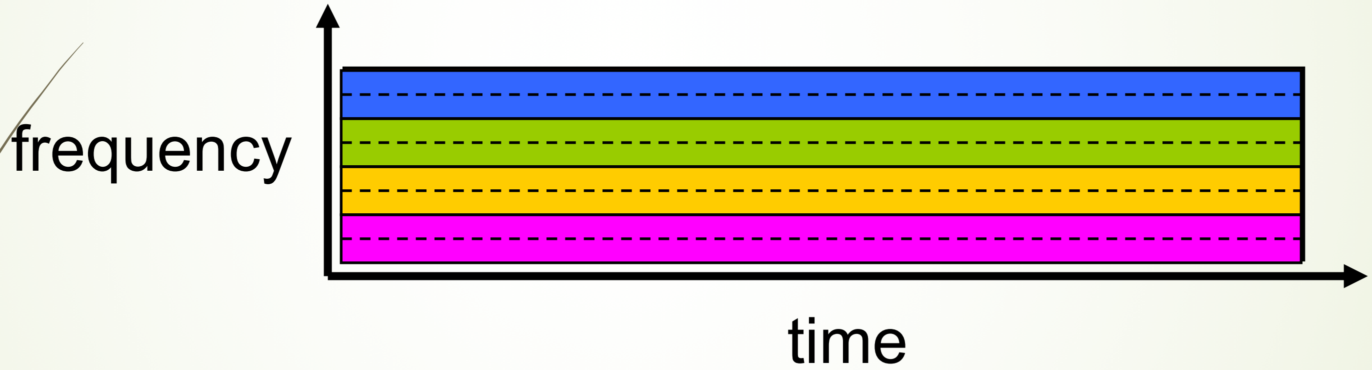


How Circuit Switching Multiplexing Among Hosts

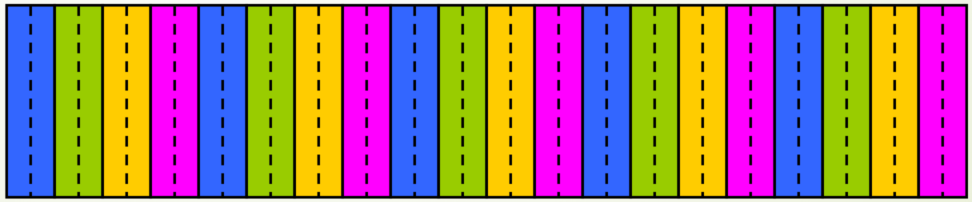
Example:

4 users

FDM



TDM

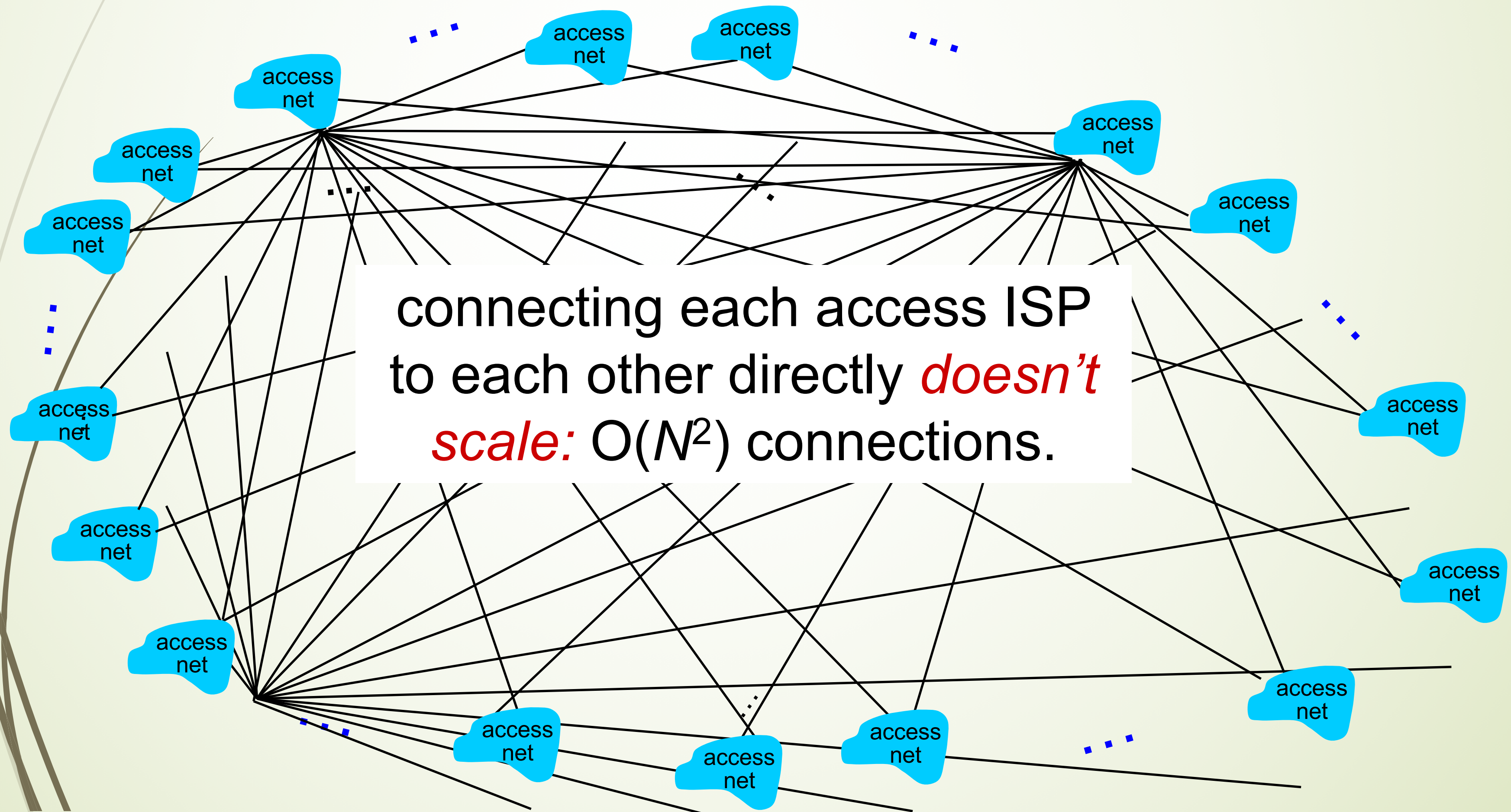


Packet Switching Versus Circuit Switching

- Q: human analogies of reserved resources (circuit switching) versus on-demand allocation (packet-switching)?
- Packet switching is great for bursty data
  - resource sharing
  - simpler, no call setup
- But it suffers from packet delay and loss
  - protocols needed for reliable data transfer, congestion control
- Q: How to provide circuit-like behavior?
  - bandwidth guarantees needed for audio/video apps
  - open problems ← will be discussed in weeks 3 and 4

Core Network Structure: Network of Networks

Challenge: given *millions* of access ISPs, how to connect them together?



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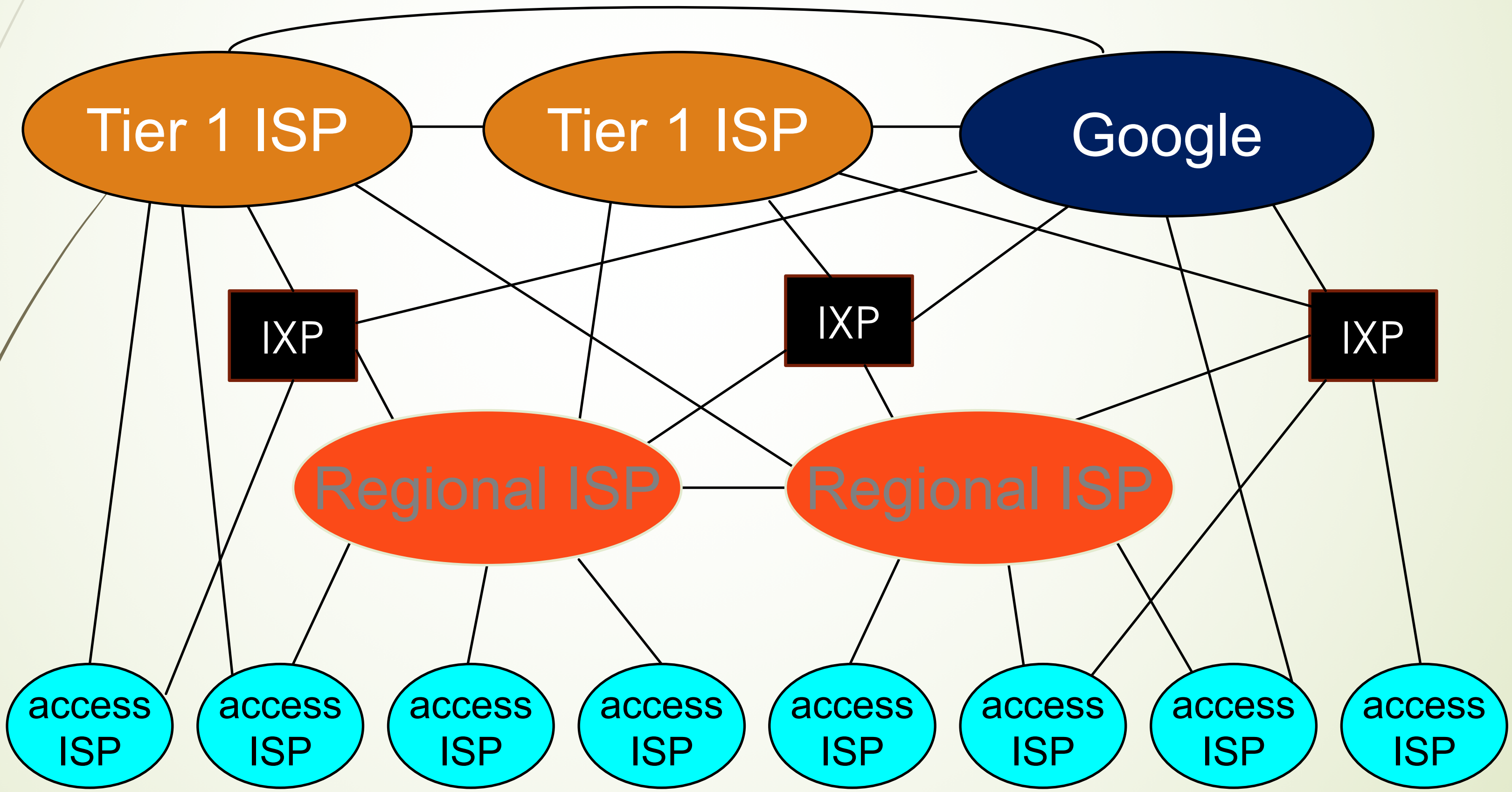
Current Core Network Structure



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How to Deal With the Excessive Complexity

- Too many pieces:
  - hosts, routers, links of various physical media, applications, protocols, hardware, software
- Layering approach: each layer implements a service
  - via internal-layer actions
  - relying on services provided by the layer below
- Benefits: Ease of discussion and system updates

Internet Protocol Stack

- **application**: supporting network applications
  - FTP, SMTP, HTTP
- **transport**: process-process data transfer
  - TCP, UDP
- **network**: routing of datagrams from source to destination hosts
  - IP, routing protocols
- **link**: data transfer between neighboring network elements
  - Ethernet, 802.111 (WiFi), PPP
- **physical**: bits on the wire